

**Question 1.2** Now let's plot Cobb-Douglas as a function of the capital stock, holding labor constant at  $\bar{L}$ . To do this, we'll need to write a function called `cobb_douglas_plotter_K` that takes in the Cobb-Douglas model parameters to create an array of values for  $Y$  called `Y_s`. In the function body, fill in the correct values for `K_s` and `Y_s`. `K_s` should be an array of the possible values of  $K$  in the range  $[0, 1]$  using intervals of 0.01. When you have completed the function, call it using  $A = 1$ ,  $\bar{L} = 0.5$ , and  $\alpha = 0.4$ .

*Hint:* Your `cobb_douglas` function can take any of its parameters as an array.

```
In [ ]: def cobb_douglas_plotter_K(A, L_bar, alpha):
        K_s = ...
        Y_s = ...

        ### Do not edit the code below; this does the plotting ###
        plt.plot(K_s, Y_s)
        plt.title(fr"Cobb-Douglas with $\bar{L}$ = {L_bar}, $A$ = {A} and $\alpha$ = {alpha}")
        plt.xlabel("Capital Stock")
        plt.ylabel("Output");

        ...
```



**Question 1.3** Suppose a country doubles its capital stock while holding labor constant. Using the Cobb-Douglas production function

$$Y = AK^\alpha L^{1-\alpha}$$

explain why output increases but at a diminishing rate. Support your answer by discussing the behavior of the marginal product of capital (MPK).

*Type your answer here, replacing this text.*



**Question 1.4.** Suppose a technological innovation doubles total factor productivity ( $A$ ). Explain how this shock affects both (i) the output level for a given  $K$  and  $L$ , and (ii) the marginal products of capital and labor. Use the Cobb-Douglas function and textbook reasoning in your explanation.

*Type your answer here, replacing this text.*



**Question 2.3.** Are there missing values in `data`? What transformation could we do on the data to mitigate against this?

*Type your answer here, replacing this text.*





**Question 2.4.** Based on our discussion of the Cobb-Douglas Production Function, which columns would be most beneficial for our analysis?

*Type your answer here, replacing this text.*



**Question 2.7.** Now let's create a histogram to visualize this information! Use the `rgdpe` column from our `data` table to create a histogram with 20 equally spaced bins.

```
In [ ]: ...
```



**Question 2.8.** Based on the summary statistics you calculated above, as well as the histogram you just created, do you observe anything of particular interest? Note that there is no ‘correct’ solution here, we just expect you to reasonably justify your choice.

*Type your answer here, replacing this text.*



**Question 2.10.** Based solely on the summary statistics you calculated, do you observe anything of particular interest? Do any of the similarities or differences have anything to do with the metric we are exploring, **emp**? Note that there is no ‘correct’ solution here, we just expect you to reasonably justify your choice.

*Type your answer here, replacing this text.*

